

1.5 ▲ Limits at Infinity

Up until this point we have discussed what happens to a function as we move its input x closer and closer to a particular point a . For a great many applications of limits we need to understand what happens to a function when its input becomes extremely large — for example what happens to a population at a time far in the future.

The definition of a limit at infinity has a similar flavour to the definition of limits at finite points that we saw above, but the details are a little different. We also need to distinguish between positive and negative infinity. As x becomes very large and positive it moves off towards $+\infty$ but when it becomes very large and negative it moves off towards $-\infty$.

Again we give an informal definition; the full formal definition can be found in (the optional) Section [1.8](#) near the end of this chapter.

Definition 1.5.1 (Limits at infinity — informal).

We write

$$\lim_{x \rightarrow \infty} f(x) = L$$

when the value of the function $f(x)$ gets closer and closer to L as we make x larger and larger and positive.

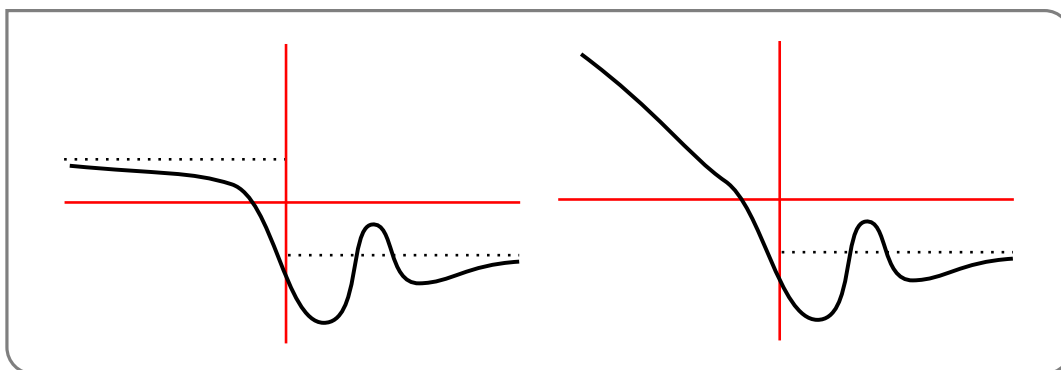
Similarly we write

$$\lim_{x \rightarrow -\infty} f(x) = L$$

when the value of the function $f(x)$ gets closer and closer to L as we make x larger and larger and negative.

Example 1.5.2

Consider the two functions depicted below



The dotted horizontal lines indicate the behaviour as x becomes very large. The function on the left has limits as $x \rightarrow \infty$ and as $x \rightarrow -\infty$ since the function “settles down” to a

particular value. On the other hand, the function on the right does not have a limit as $x \rightarrow -\infty$ since the function just keeps getting bigger and bigger.

Example 1.5.2

Just as was the case for limits as $x \rightarrow a$ we will start with two very simple building blocks and build other limits from those.

Theorem 1.5.3.

Let $c \in \mathbb{R}$ then the following limits hold

$$\lim_{x \rightarrow \infty} c = c$$

$$\lim_{x \rightarrow -\infty} c = c$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

Again, these limits interact nicely with standard arithmetic:

Theorem 1.5.4 (Arithmetic of limits at infinity).

Let $f(x), g(x)$ be two functions for which the limits

$$\lim_{x \rightarrow \infty} f(x) = F$$

$$\lim_{x \rightarrow \infty} g(x) = G$$

exist. Then the following limits hold

$$\lim_{x \rightarrow \infty} f(x) \pm g(x) = F \pm G$$

$$\lim_{x \rightarrow \infty} f(x)g(x) = FG$$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{F}{G} \quad \text{provided } G \neq 0$$

and for real numbers p

$$\lim_{x \rightarrow \infty} f(x)^p = F^p \quad \text{provided } F^p \text{ and } f(x)^p \text{ are defined for all } x$$

The analogous results hold for limits to $-\infty$.

Note that, as was the case in Theorem [1.4.8](#), we need a little extra care with powers of functions. We must avoid taking square roots of negative numbers, or indeed any even root of a negative number^{[21](#)}.

²¹ To be more precise, there is no real number x so that $x^{\text{even power}}$ is a negative number. Hence we cannot take the even-root of a negative number and express it as a real number. This is precisely what complex numbers allow us to do, but alas there is not space in the course for us to explore them.

Hence we have for all rational $r > 0$

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

but we have to be careful with

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$$

This is only true if the denominator of r is not an even number²².

For example

- $\lim_{x \rightarrow \infty} \frac{1}{x^{1/2}} = 0$, but $\lim_{x \rightarrow -\infty} \frac{1}{x^{1/2}}$ does not exist, because $x^{1/2}$ is not defined for $x < 0$.
- On the other hand, $x^{4/3}$ is defined for negative values of x and $\lim_{x \rightarrow -\infty} \frac{1}{x^{4/3}} = 0$.

Our first application of limits at infinity will be to examine the behaviour of a rational function for very large x . To do this we use a “trick”.

Example 1.5.5

Compute the following limit:

$$\lim_{x \rightarrow \infty} \frac{x^2 - 3x + 4}{3x^2 + 8x + 1}$$

As x becomes very large, it is the x^2 term that will dominate in both the numerator and denominator and the other bits become irrelevant. That is, for very large x , x^2 is much much larger than x or any constant. So we pull out these dominant parts

$$\begin{aligned} \frac{x^2 - 3x + 4}{3x^2 + 8x + 1} &= \frac{x^2 \left(1 - \frac{3}{x} + \frac{4}{x^2}\right)}{x^2 \left(3 + \frac{8}{x} + \frac{1}{x^2}\right)} \\ &= \frac{1 - \frac{3}{x} + \frac{4}{x^2}}{3 + \frac{8}{x} + \frac{1}{x^2}} \end{aligned} \quad \text{remove the common factors}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^2 - 3x + 4}{3x^2 + 8x + 1} &= \lim_{x \rightarrow \infty} \frac{1 - \frac{3}{x} + \frac{4}{x^2}}{3 + \frac{8}{x} + \frac{1}{x^2}} \\ &= \frac{\lim_{x \rightarrow \infty} \left(1 - \frac{3}{x} + \frac{4}{x^2}\right)}{\lim_{x \rightarrow \infty} \left(3 + \frac{8}{x} + \frac{1}{x^2}\right)} \end{aligned} \quad \text{arithmetic of limits}$$

$$\begin{aligned} &= \frac{\lim_{x \rightarrow \infty} 1 - \lim_{x \rightarrow \infty} \frac{3}{x} + \lim_{x \rightarrow \infty} \frac{4}{x^2}}{\lim_{x \rightarrow \infty} 3 + \lim_{x \rightarrow \infty} \frac{8}{x} + \lim_{x \rightarrow \infty} \frac{1}{x^2}} \\ &= \frac{1 + 0 + 0}{3 + 0 + 0} = \frac{1}{3} \end{aligned} \quad \text{more arithmetic of limits}$$

²² where we write $r = \frac{p}{q}$ with p, q integers with no common factors. For example, $r = \frac{6}{14}$ should be written as $r = \frac{3}{7}$ when considering this rule.

Example 1.5.5

The following one gets a little harder

Example 1.5.6

Find the limit as $x \rightarrow \infty$ of $\frac{\sqrt{4x^2+1}}{5x-1}$

We use the same trick — try to work out what is the biggest term in the numerator and denominator and pull it to one side.

- The denominator is dominated by $5x$.
- The biggest contribution to the numerator comes from the $4x^2$ inside the square-root. When we pull x^2 outside the square-root it becomes x , so the numerator is dominated by $x \cdot \sqrt{4} = 2x$
- To see this more explicitly rewrite the numerator

$$\sqrt{4x^2+1} = \sqrt{x^2(4+1/x^2)} = \sqrt{x^2}\sqrt{4+1/x^2} = x\sqrt{4+1/x^2}.$$

- Thus the limit as $x \rightarrow \infty$ is

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{4x^2+1}}{5x-1} &= \lim_{x \rightarrow \infty} \frac{x\sqrt{4+1/x^2}}{x(5-1/x)} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{4+1/x^2}}{5-1/x} \\ &= \frac{2}{5} \end{aligned}$$

Example 1.5.6

Now let us also think about the limit of the same function, $\frac{\sqrt{4x^2+1}}{5x-1}$, as $x \rightarrow -\infty$. There is something subtle going on because of the square-root. First consider the function²³

$$h(t) = \sqrt{t^2}$$

Evaluating this at $t = 7$ gives

$$h(7) = \sqrt{7^2} = \sqrt{49} = 7$$

We'll get much the same thing for any $t \geq 0$. For any $t \geq 0$, $h(t) = \sqrt{t^2}$ returns exactly t . However now consider the function at $t = -3$

$$h(-3) = \sqrt{(-3)^2} = \sqrt{9} = 3 = -(-3)$$

23 Just to change things up let's use t and $h(t)$ instead of the ubiquitous x and $f(x)$.

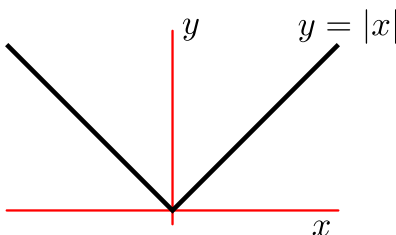
that is the function is returning -1 times the input.

This is because when we defined $\sqrt{}$, we defined it to be the *positive* square-root. i.e. the function $\sqrt{t}$ can never return a negative number. So being more careful

$$h(t) = \sqrt{t^2} = |t|$$

Where the $|t|$ is the absolute value of t . You are perhaps used to thinking of absolute value as “remove the minus sign”, but this is not quite correct. Let’s sketch the function

Figure 1.5.1.



It is a piecewise function defined by

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Hence our function $h(t)$ is really

$$h(t) = \sqrt{t^2} = \begin{cases} t & t \geq 0 \\ -t & t < 0 \end{cases}$$

So that when we evaluate $h(-7)$ it is

$$h(-7) = \sqrt{(-7)^2} = \sqrt{49} = 7 = -(-7)$$

We are now ready to examine the limit as $x \rightarrow -\infty$ in our previous example. Mostly it is copy and paste from above.

Example 1.5.7

Find the limit as $x \rightarrow -\infty$ of $\frac{\sqrt{4x^2+1}}{5x-1}$

We use the same trick — try to work out what is the biggest term in the numerator and denominator and pull it to one side. Since we are taking the limit as $x \rightarrow -\infty$ we should think of x as a large negative number.

- The denominator is dominated by $5x$.
- The biggest contribution to the numerator comes from the $4x^2$ inside the square-root. When we pull the x^2 outside a square-root it becomes $|x| = -x$ (since we are taking the limit as $x \rightarrow -\infty$), so the numerator is dominated by $-x \cdot \sqrt{4} = -2x$

- To see this more explicitly rewrite the numerator

$$\begin{aligned}\sqrt{4x^2 + 1} &= \sqrt{x^2(4 + 1/x^2)} = \sqrt{x^2}\sqrt{4 + 1/x^2} \\ &= |x|\sqrt{4 + 1/x^2} \quad \text{and since } x < 0 \text{ we have} \\ &= -x\sqrt{4 + 1/x^2}\end{aligned}$$

- Thus the limit as $x \rightarrow -\infty$ is

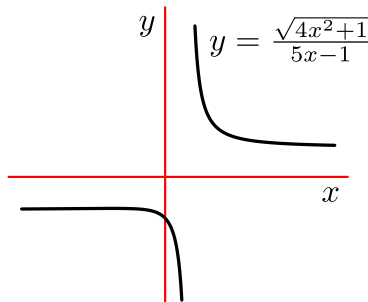
$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 1}}{5x - 1} &= \lim_{x \rightarrow -\infty} \frac{-x\sqrt{4 + 1/x^2}}{x(5 - 1/x)} \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt{4 + 1/x^2}}{5 - 1/x} \\ &= -\frac{2}{5}\end{aligned}$$

Example 1.5.7

So the limit as $x \rightarrow -\infty$ is almost the same but we gain a minus sign. This **is definitely not** the case in general — you have to think about each example separately.

Here is a sketch of the function in question.

Figure 1.5.2.



Example 1.5.8

Compute the following limit:

$$\lim_{x \rightarrow \infty} (x^{7/5} - x)$$

In this case we cannot use the arithmetic of limits to write this as

$$\begin{aligned}\lim_{x \rightarrow \infty} (x^{7/5} - x) &= \left(\lim_{x \rightarrow \infty} x^{7/5} \right) - \left(\lim_{x \rightarrow \infty} x \right) \\ &= \infty - \infty\end{aligned}$$

because the limits do not exist. We can only use the limit laws when the limits exist. So we should go back and think some more.

When x is very large, $x^{7/5} = x \cdot x^{2/5}$ will be much larger than x , so the $x^{7/5}$ term will dominate the x term. So factor out $x^{7/5}$ and rewrite it as

$$x^{7/5} - x = x^{7/5} \left(1 - \frac{1}{x^{2/5}} \right)$$

Consider what happens to each of the factors as $x \rightarrow \infty$

- For large x , $x^{7/5} > x$ (this is actually true for any $x > 1$). In the limit as $x \rightarrow +\infty$, x becomes arbitrarily large and positive, and $x^{7/5}$ must be bigger still, so it follows that

$$\lim_{x \rightarrow \infty} x^{7/5} = +\infty.$$

- On the other hand, $(1 - x^{-2/5})$ becomes closer and closer to 1 — we can use the arithmetic of limits to write this as

$$\lim_{x \rightarrow \infty} (1 - x^{-2/5}) = \lim_{x \rightarrow \infty} 1 - \lim_{x \rightarrow \infty} x^{-2/5} = 1 - 0 = 1$$

So the product of these two factors will be come larger and larger (and positive) as x moves off to infinity. Hence we have

$$\lim_{x \rightarrow \infty} x^{7/5} \left(1 - 1/x^{2/5} \right) = +\infty$$

Example 1.5.8

But remember $+\infty$ and $-\infty$ are not numbers; the last equation in the example is shorthand for “the function becomes arbitrarily large”.

In the previous section we saw that finite limits and arithmetic interact very nicely (see Theorems [1.4.2](#) and [1.4.8](#)). This enabled us to compute the limits of more complicated function in terms of simpler ones. When limits of functions go to plus or minus infinity we are quite a bit more restricted in what we can deduce. The next theorem states some results concerning the sum, difference, ratio and product of infinite limits — unfortunately in many cases we cannot make general statements and the results will depend on the details of the problem at hand.

Theorem 1.5.9 (Arithmetic of infinite limits).

Let $a, c, H \in \mathbb{R}$ and let f, g, h be functions defined in an interval around a (but they need not be defined at $x = a$), so that

$$\lim_{x \rightarrow a} f(x) = +\infty$$

$$\lim_{x \rightarrow a} g(x) = +\infty$$

$$\lim_{x \rightarrow a} h(x) = H$$

- $\lim_{x \rightarrow a} (f(x) + g(x)) = +\infty$
- $\lim_{x \rightarrow a} (f(x) + h(x)) = +\infty$
- $\lim_{x \rightarrow a} (f(x) - g(x))$ undetermined
- $\lim_{x \rightarrow a} (f(x) - h(x)) = +\infty$
- $\lim_{x \rightarrow a} cf(x) = \begin{cases} +\infty & c > 0 \\ 0 & c = 0 \\ -\infty & c < 0 \end{cases}$
- $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = +\infty$.
- $\lim_{x \rightarrow a} f(x)h(x) = \begin{cases} +\infty & H > 0 \\ -\infty & H < 0 \\ \text{undetermined} & H = 0 \end{cases}$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ undetermined
- $\lim_{x \rightarrow a} \frac{f(x)}{h(x)} = \begin{cases} +\infty & H > 0 \\ -\infty & H < 0 \\ \text{undetermined} & H = 0 \end{cases}$
- $\lim_{x \rightarrow a} \frac{h(x)}{f(x)} = 0$
- $\lim_{x \rightarrow a} f(x)^p = \begin{cases} +\infty & p > 0 \\ 0 & p < 0 \\ 1 & p = 0 \end{cases}$

Note that by “undetermined” we mean that the limit may or may not exist, but cannot be determined from the information given in the theorem. See Example [1.4.6](#) for an example of what we mean by “undetermined”. Additionally consider the following example.

Example 1.5.10

Consider the following 3 functions:

$$f(x) = x^{-2}$$

$$g(x) = 2x^{-2}$$

$$h(x) = x^{-2} - 1.$$

Their limits as $x \rightarrow 0$ are:

$$\lim_{x \rightarrow 0} f(x) = +\infty$$

$$\lim_{x \rightarrow 0} g(x) = +\infty$$

$$\lim_{x \rightarrow 0} h(x) = +\infty.$$

Say we want to compute the limit of the difference of two of the above functions as $x \rightarrow 0$. Then the previous theorem cannot help us. This is not because it is too weak, rather it is because the difference of two infinite limits can be, either plus infinity, minus infinity or some finite number depending on the details of the problem. For example,

$$\lim_{x \rightarrow 0} (f(x) - g(x)) = \lim_{x \rightarrow 0} -x^{-2} = -\infty$$

$$\lim_{x \rightarrow 0} (f(x) - h(x)) = \lim_{x \rightarrow 0} 1 = 1$$

$$\lim_{x \rightarrow 0} (g(x) - h(x)) = \lim_{x \rightarrow 0} x^{-2} + 1 = +\infty$$

Example 1.5.10

1.6 ▲ Continuity

We have seen that computing the limits some functions — polynomials and rational functions — is very easy because

$$\lim_{x \rightarrow a} f(x) = f(a).$$

That is, the limit as x approaches a is just $f(a)$. Roughly speaking, the reason we can compute the limit this way is that these functions do not have any abrupt jumps near a .

Many other functions have this property, $\sin(x)$ for example. A function with this property is called “continuous” and there is a precise mathematical definition for it. If you do not recall interval notation, then now is a good time to take a quick look back at Definition [0.3.5](#).

Definition 1.6.1.

A function $f(x)$ is continuous at a if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

If a function is not continuous at a then it is said to be discontinuous at a .

When we write that f is continuous without specifying a point, then typically this means that f is continuous at a for all $a \in \mathbb{R}$.

When we write that $f(x)$ is continuous on the open interval (a, b) then the function is continuous at every point c satisfying $a < c < b$.

So if a function is continuous at $x = a$ we immediately know that

- $f(a)$ exists
- $\lim_{x \rightarrow a^-}$ exists and is equal to $f(a)$, and
- $\lim_{x \rightarrow a^+}$ exists and is equal to $f(a)$.

► Quick Aside — One-sided Continuity

Notice in the above definition of continuity on an interval (a, b) we have carefully avoided saying anything about whether or not the function is continuous at the endpoints of the interval — i.e. is $f(x)$ continuous at $x = a$ or $x = b$. This is because talking of continuity at the endpoints of an interval can be a little delicate.

In many situations we will be given a function $f(x)$ defined on a closed interval $[a, b]$. For example, we might have:

$$f(x) = \frac{x+1}{x+2} \quad \text{for } x \in [0, 1].$$

For any $0 \leq x \leq 1$ we know the value of $f(x)$. However for $x < 0$ or $x > 1$ we know nothing about the function — indeed it has not been defined.

So now, consider what it means for $f(x)$ to be continuous at $x = 0$. We need to have

$$\lim_{x \rightarrow 0} f(x) = f(0),$$

however this implies that the one-sided limits

$$\lim_{x \rightarrow 0^+} f(x) = f(0) \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = f(0)$$

Now the first of these one-sided limits involves examining the behaviour of $f(x)$ for $x > 0$. Since this involves looking at points for which $f(x)$ is defined, this is something we can do. On the other hand the second one-sided limit requires us to understand the behaviour of $f(x)$ for $x < 0$. This we cannot do because the function hasn't been defined for $x < 0$.

One way around this problem is to generalise the idea of continuity to one-sided continuity, just as we generalised limits to get one-sided limits.

Definition 1.6.2.

A function $f(x)$ is continuous from the right at a if

$$\lim_{x \rightarrow a^+} f(x) = f(a).$$

Similarly a function $f(x)$ is continuous from the left at a if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

Using the definition of one-sided continuity we can now define what it means for a function to be continuous on a closed interval.

Definition 1.6.3.

A function $f(x)$ is continuous on the closed interval $[a, b]$ when

- $f(x)$ is continuous on (a, b) ,
- $f(x)$ is continuous from the right at a , and
- $f(x)$ is continuous from the left at b .

Note that the last two conditions are equivalent to

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow b^-} f(x) = f(b).$$

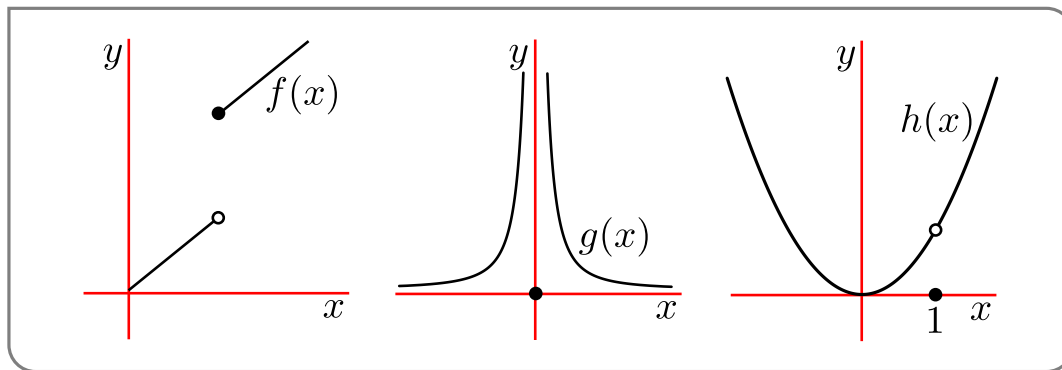
► **Back to the Main Text**

We already know from our work above that polynomials are continuous, and that rational functions are continuous at all points in their domains — i.e. where their denominators are non-zero. As we did for limits, we will see that continuity interacts “nicely” with arithmetic. This will allow us to construct complicated continuous functions from simpler continuous building blocks (like polynomials).

But first, a few examples...

Example 1.6.4

Consider the functions drawn below



These are

$$f(x) = \begin{cases} x & x < 1 \\ x + 2 & x \geq 1 \end{cases} \quad g(x) = \begin{cases} 1/x^2 & x \neq 0 \\ 0 & x = 0 \end{cases} \quad h(x) = \begin{cases} \frac{x^3 - x^2}{x - 1} & x \neq 1 \\ 0 & x = 1 \end{cases}$$

Determine where they are continuous and discontinuous:

- When $x < 1$ then $f(x)$ is a straight line (and so a polynomial) and so it is continuous at every point $x < 1$. Similarly when $x > 1$ the function is a straight line and so it is

continuous at every point $x > 1$. The only point which might be a discontinuity is at $x = 1$. We see that the one sided limits are different. Hence the limit at $x = 1$ does not exist and so the function is discontinuous at $x = 1$.

But note that that $f(x)$ is continuous from one side — which?

- The middle case is much like the previous one. When $x \neq 0$ the $g(x)$ is a rational function and so is continuous everywhere on its domain (which is all reals except $x = 0$). Thus the only point where $g(x)$ might be discontinuous is at $x = 0$. We see that neither of the one-sided limits exist at $x = 0$, so the limit does not exist at $x = 0$. Hence the function is discontinuous at $x = 0$.
- We have seen the function $h(x)$ before. By the same reasoning as above, we know it is continuous except at $x = 1$ which we must check separately.

By definition of $h(x)$, $h(1) = 0$. We must compare this to the limit as $x \rightarrow 1$. We did this before.

$$\frac{x^3 - x^2}{x - 1} = \frac{x^2(x - 1)}{x - 1} = x^2$$

So $\lim_{x \rightarrow 1} \frac{x^3 - x^2}{x - 1} = \lim_{x \rightarrow 1} x^2 = 1 \neq h(1)$. Hence h is discontinuous at $x = 1$.

Example 1.6.4

This example illustrates different sorts of discontinuities:

- The function $f(x)$ has a “jump discontinuity” because the function “jumps” from one finite value on the left to another value on the right.
- The second function, $g(x)$, has an “infinite discontinuity” since $\lim f(x) = +\infty$.
- The third function, $h(x)$, has a “removable discontinuity” because we could make the function continuous at that point by redefining the function at that point. i.e. setting $h(1) = 1$. That is

$$\text{new function } h(x) = \begin{cases} \frac{x^3 - x^2}{x - 1} & x \neq 1 \\ 1 & x = 1 \end{cases}$$

Showing a function is continuous can be a pain, but just as the limit laws help us compute complicated limits in terms of simpler limits, we can use them to show that complicated functions are continuous by breaking them into simpler pieces.

Theorem 1.6.5 (Arithmetic of continuity).

Let $a, c \in \mathbb{R}$ and let $f(x)$ and $g(x)$ be functions that are continuous at a . Then the following functions are also continuous at $x = a$:

- $f(x) + g(x)$ and $f(x) - g(x)$,
- $cf(x)$ and $f(x)g(x)$, and
- $\frac{f(x)}{g(x)}$ provided $g(a) \neq 0$.

Above we stated that polynomials and rational functions are continuous (being careful about domains of rational functions — we must avoid the denominators being zero) without making it a formal statement. This is easily fixed...

Lemma 1.6.6.

Let $c \in \mathbb{R}$. The functions

$$f(x) = x$$

$$g(x) = c$$

are continuous everywhere on the real line

This isn't quite the result we wanted (that's a couple of lines below) but it is a small result that we can combine with the arithmetic of limits to get the result we want. Such small helpful results are called "lemmas" and they will arise more as we go along.

Now since we can obtain any polynomial and any rational function by carefully adding, subtracting, multiplying and dividing the functions $f(x) = x$ and $g(x) = c$, the above lemma combines with the "arithmetic of continuity" theorem to give us the result we want:

Theorem 1.6.7 (Continuity of polynomials and rational functions).

Every polynomial is continuous everywhere. Similarly every rational function is continuous except where its denominator is zero (i.e. on all its domain).

With some more work this result can be extended to wider families of functions:

Theorem 1.6.8.

The following functions are continuous everywhere in their domains

- polynomials, rational functions
- roots and powers
- trig functions and their inverses
- exponential and the logarithm

We haven't encountered inverse trigonometric functions, nor exponential functions or logarithms, but we will see them in the next chapter. For the moment, just file the information away.

Using a combination of the above results you can show that many complicated functions are continuous except at a few points (usually where a denominator is equal to zero).

Example 1.6.9

Where is the function $f(x) = \frac{\sin(x)}{2+\cos(x)}$ continuous?

We just break things down into pieces and then put them back together keeping track of where things might go wrong.

- The function is a ratio of two pieces — so check if the numerator is continuous, the denominator is continuous, and if the denominator might be zero.
- The numerator is $\sin(x)$ which is “continuous on its domain” according to one of the above theorems. Its domain is all real numbers²⁴ so it is continuous everywhere. No problems here.
- The denominator is the sum of 2 and $\cos(x)$. Since 2 is a constant it is continuous everywhere. Similarly (we just checked things for the previous point) we know that $\cos(x)$ is continuous everywhere. Hence the denominator is continuous.
- So we just need to check if the denominator is zero. One of the facts that we should know²⁵ is that

$$-1 \leq \cos(x) \leq 1$$

and so by adding 2 we get

$$1 \leq 2 + \cos(x) \leq 3$$

Thus no matter what value of x , $2 + \cos(x) \geq 1$ and so cannot be zero.

24 Remember that $\sin$ and $\cos$ are defined on all real numbers, so $\tan(x) = \sin(x)/\cos(x)$ is continuous everywhere except where $\cos(x) = 0$. This happens when $x = \frac{\pi}{2} + n\pi$ for any integer n . If you cannot remember where $\tan(x)$ “blows up” or $\sin(x) = 0$ or $\cos(x) = 0$ then you should definitely revise trigonometric functions. Come to think of it — just revise them anyway.

25 If you do not know this fact then you should revise trigonometric functions. See the previous footnote.

- So the numerator is continuous, the denominator is continuous and nowhere zero, so the function is continuous everywhere.

If the function were changed to $\frac{\sin(x)}{x^2 - 5x + 6}$ much of the same reasoning can be used. Being a little terse we could answer with:

- Numerator and denominator are continuous.
- Since $x^2 - 5x + 6 = (x - 2)(x - 3)$ the denominator is zero when $x = 2, 3$.
- So the function is continuous everywhere except possibly at $x = 2, 3$. In order to verify that the function really is discontinuous at those points, it suffices to verify that the numerator is non-zero at $x = 2, 3$. Indeed we know that $\sin(x)$ is zero only when $x = n\pi$ (for any integer n). Hence $\sin(2), \sin(3) \neq 0$. Thus the numerator is non-zero, while the denominator is zero and hence $x = 2, 3$ really are points of discontinuity.

Note that this example raises a subtle point about checking continuity when numerator and denominator are *simultaneously* zero. There are quite a few possible outcomes in this case and we need more sophisticated tools to adequately analyse the behaviour of functions near such points. We will return to this question later in the text after we have developed Taylor expansions (see Section 3.4).

Example 1.6.9

So we know what happens when we add subtract multiply and divide, what about when we compose functions? Well - limits and compositions work nicely when things are continuous.

Theorem 1.6.10 (Compositions and continuity).

If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$ then $\lim_{x \rightarrow a} f(g(x)) = f(b)$. I.e.

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

Hence if g is continuous at a and f is continuous at $g(a)$ then the composite function $(f \circ g)(x) = f(g(x))$ is continuous at a .

So when we compose two continuous functions we get a new continuous function. We can put this to use

Example 1.6.11

Where are the following functions continuous?

$$f(x) = \sin(x^2 + \cos(x))$$

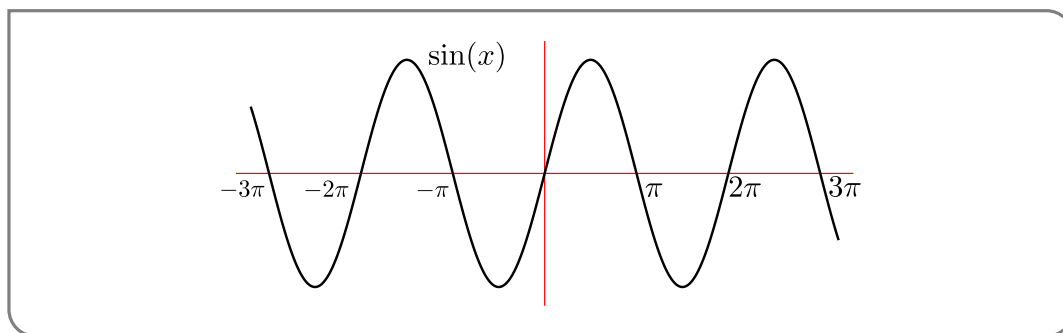
$$g(x) = \sqrt{\sin(x)}$$

Our first step should be to break the functions down into pieces and study them. When we put them back together we should be careful of dividing by zero, or falling outside the domain.

- The function $f(x)$ is the composition of $\sin(x)$ with $x^2 + \cos(x)$.
- These pieces, $\sin(x)$, x^2 , $\cos(x)$ are continuous everywhere.
- So the sum $x^2 + \cos(x)$ is continuous everywhere
- And hence the composition of $\sin(x)$ and $x^2 + \cos(x)$ is continuous everywhere.

The second function is a little trickier.

- The function $g(x)$ is the composition of $\sqrt{x}$ with $\sin(x)$.
- $\sqrt{x}$ is continuous on its domain $x \geq 0$.
- $\sin(x)$ is continuous everywhere, but it is negative in many places.
- In order for $g(x)$ to be defined and continuous we must restrict x so that $\sin(x) \geq 0$.
- Recall the graph of $\sin(x)$:



Hence $\sin(x) \geq 0$ when $x \in [0, \pi]$ or $x \in [2\pi, 3\pi]$ or $x \in [-2\pi, -\pi]$ or ... To be more precise $\sin(x)$ is positive when $x \in [2n\pi, (2n+1)\pi]$ for any integer n .

- Hence $g(x)$ is continuous when $x \in [2n\pi, (2n+1)\pi]$ for any integer n .

Example 1.6.11

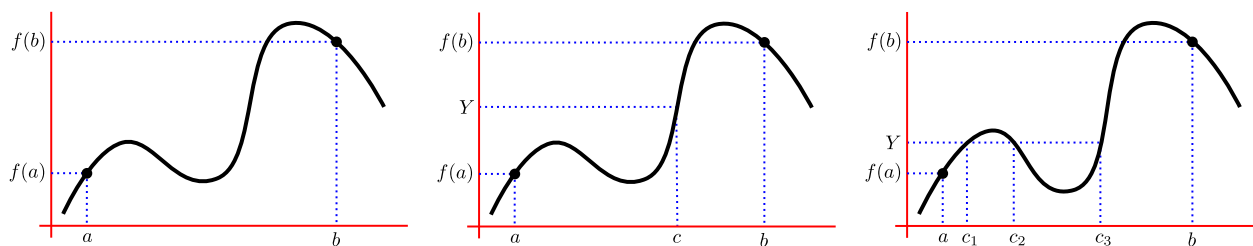
Continuous functions are very nice (mathematically speaking). Functions from the “real world” tend to be continuous (though not always). The key aspect that makes them nice is the fact that they don’t jump about.

The absence of such jumps leads to the following theorem which, while it can be quite confusing on first glance, actually says something very natural — obvious even. It says, roughly speaking, that, as you draw the graph $y = f(x)$ starting at $x = a$ and ending at $x = b$, y changes continuously from $y = f(a)$ to $y = f(b)$, with no jumps, and consequently y must take every value between $f(a)$ and $f(b)$ at least once. We’ll start by just giving the precise statement and then we’ll explain it in detail.

Theorem 1.6.12 (Intermediate value theorem (IVT)).

Let $a < b$ and let f be a function that is continuous at all points $a \leq x \leq b$. If Y is any number between $f(a)$ and $f(b)$ then there exists some number $c \in [a, b]$ so that $f(c) = Y$.

Like the $\epsilon - \delta$ definition of limits²⁶, we should break this theorem down into pieces. Before we do that, keep the following pictures in mind.

Figure 1.6.1.

Now the break-down

- **Let $a < b$ and let f be a function that is continuous at all points $a \leq x \leq b$.** — This is setting the scene. We have a, b with $a < b$ (we can safely assume these to be real numbers). Our function must be continuous at all points between a and b .
- **if Y is any number between $f(a)$ and $f(b)$** — Now we need another number Y and the only restriction on it is that it lies between $f(a)$ and $f(b)$. That is, if $f(a) \leq f(b)$ then $f(a) \leq Y \leq f(b)$. Or if $f(a) \geq f(b)$ then $f(a) \geq Y \geq f(b)$. So notice that Y could be equal to $f(a)$ or $f(b)$ — if we wanted to avoid that possibility, then we would normally explicitly say $Y \neq f(a), f(b)$ or we would write that Y is *strictly* between $f(a)$ and $f(b)$.
- **there exists some number $c \in [a, b]$ so that $f(c) = Y$** — so if we satisfy all of the above conditions, then there has to be some real number c lying between a and b so that when we evaluate $f(c)$ it is Y .

So that breaks down the theorem statement by statement, but what does it actually mean?

- Draw any continuous function you like between a and b — it must be continuous.
- The function takes the value $f(a)$ at $x = a$ and $f(b)$ at $x = b$ — see the left-hand figure above.

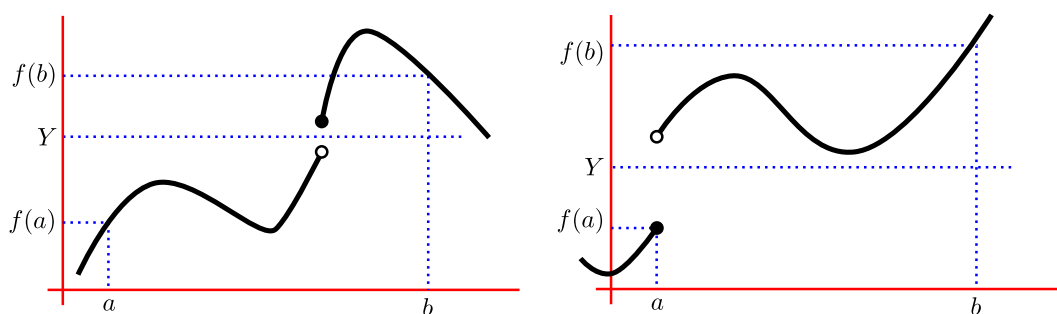
²⁶ The interested student is invited to take a look at the optional Section [1.7](#)

- Now we can pick any Y that lies between $f(a)$ and $f(b)$ — see the middle figure above. The IVT²⁷ tells us that there must be some x -value that when plugged into the function gives us Y . That is, there is some c between a and b so that $f(c) = Y$. We can also interpret this graphically; the IVT tells us that the horizontal straight line $y = Y$ must intersect the graph $y = f(x)$ at some point (c, Y) with $a \leq c \leq b$.
- Notice that the IVT does not tell us how many such c -values there are, just that there is at least one of them. See the right-hand figure above. For that particular choice of Y there are three different c values so that $f(c_1) = f(c_2) = f(c_3) = Y$.

This theorem says that if $f(x)$ is a continuous function on all of the interval $a \leq x \leq b$ then as x moves from a to b , $f(x)$ takes every value between $f(a)$ and $f(b)$ at least once. To put this slightly differently, if f were to avoid a value between $f(a)$ and $f(b)$ then f cannot be continuous on $[a, b]$.

It is not hard to convince yourself that the continuity of f is crucial to the IVT. Without it one can quickly construct examples of functions that contradict the theorem. See the figure below for a few non-continuous examples:

Figure 1.6.2.



In the left-hand example we see that a discontinuous function can “jump” over the Y -value we have chosen, so there is no x -value that makes $f(x) = Y$. The right-hand example demonstrates why we need to be careful with the ends of the interval. In particular, a function must be continuous over the whole interval $[a, b]$ *including* the end-points of the interval. If we only required the function to be continuous on (a, b) (so strictly between a and b) then the function could “jump” over the Y -value at a or b .

If you are still confused then here is a “real-world” example

Example 1.6.13

You are climbing the Grouse-grind²⁸ with a friend — call him Bob. Bob was eager and

²⁷ Often with big important useful theorems like this one, writing out the full name again and again becomes tedious, so we abbreviate it. Such abbreviations are okay provided the reader knows this is what you are doing, so the first time you use an abbreviation you should let the reader know. Much like we are doing here in this footnote: “IVT” stands for “intermediate value theorem”, which is Theorem 1.6.12.

²⁸ If you don’t know it then google it.

started at 9am. Bob, while very eager, is also very clumsy; he sprained his ankle somewhere along the path and has stopped moving at 9:21am and is just sitting²⁹ enjoying the view. You get there late and start climbing at 10am and being quite fit you get to the top at 11am. The IVT implies that at some time between 10am and 11am you meet up with Bob.

You can translate this situation into the form of the IVT as follows. Let t be time and let $a = 10\text{am}$ and $b = 11\text{am}$. Let $g(t)$ be your distance along the trail. Hence³⁰ $g(a) = 0$ and $g(b) = 2.9\text{km}$. Since you are a mortal, your position along the trail is a continuous function — no helicopters or teleportation or... We have no idea where Bob is sitting, except that he is somewhere between $g(a)$ and $g(b)$, call this point Y . The IVT guarantees that there is some time c between a and b (so between 10am and 11am) with $g(c) = Y$ (and your position will be the same as Bob's).

Example 1.6.13

Aside from finding Bob sitting by the side of the trail, one of the most important applications of the IVT is determining where a function is zero. For quadratics we know (or should know) that

$$ax^2 + bx + c = 0 \quad \text{when } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

While the Babylonians could (mostly, but not quite) do the above, the corresponding formula for solving a cubic is uglier and that for a quartic is uglier still. One of the most famous results in mathematics demonstrates that no such formula exists for quintics or higher degree polynomials³¹.

So even for polynomials we cannot, in general, write down explicit formulae for their zeros and have to make do with numerical approximations — i.e. write down the root as a decimal expansion to whatever precision we desire. For more complicated functions we have no choice — there is no reason that the zeros should be expressible as nice neat little formulas. At the same time, finding the zeros of a function:

$$f(x) = 0$$

or solving equations of the form³²

$$g(x) = h(x)$$

can be a crucial step in many mathematical proofs and applications.

29 Hopefully he remembered to carry something warm.

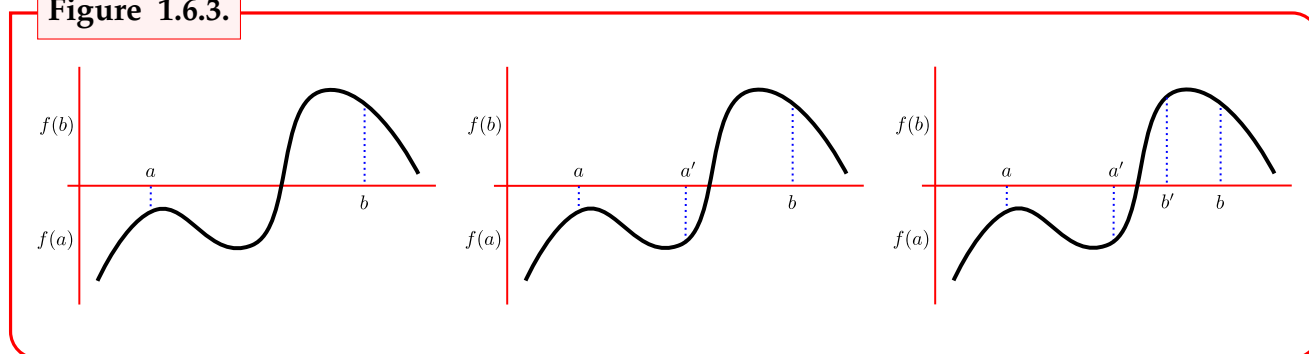
30 It's amazing what facts you can find on Wikipedia.

31 The similar (but uglier) formula for solving cubics took until the 15th century and the work of del Ferro and Cardano (and Cardano's student Ferrari). A similar (but even uglier) formula for quartics was also found by Ferrari. The extremely famous Abel-Ruffini Theorem (nearly by Ruffini in the late 18th century and completely by Abel in early 19th century) demonstrates that a similar formula for the zeros of a quintic does not exist. Note that the theorem does *not* say that quintics do not have zeros; rather it says that the zeros cannot in general be expressed using a finite combination of addition, multiplication, division, powers and roots. The interested student should also look up Évariste Galois and his contributions to this area.

32 In fact both of these are the same because we can write $f(x) = g(x) - h(x)$ and then the zeros of $f(x)$ are exactly when $g(x) = h(x)$.

For this reason there is a considerable body of mathematics which focuses just on finding the zeros of functions. The IVT provides a very simple way to “locate” the zeros of a function. In particular, if we know a continuous function is negative at a point $x = a$ and positive at another point $x = b$, then there must (by the IVT) be a point $x = c$ between a and b where $f(c) = 0$.

Figure 1.6.3.



Consider the leftmost of the above figures. It depicts a continuous function that is negative at $x = a$ and positive at $x = b$. So choose $Y = 0$ and apply the IVT — there must be some c with $a \leq c \leq b$ so that $f(c) = Y = 0$. While this doesn't tell us c exactly, it does give us bounds on the possible positions of at least one zero — there must be at least one c obeying $a \leq c \leq b$.

See middle figure. To get better bounds we could test a point half-way between a and b . So set $a' = \frac{a+b}{2}$. In this example we see that $f(a')$ is negative. Applying the IVT again tells us there is some c between a' and b so that $f(c) = 0$. Again — we don't have c exactly, but we have halved the range of values it could take.

Look at the rightmost figure and do it again — test the point half-way between a' and b . In this example we see that $f(b')$ is positive. Applying the IVT tells us that there is some c between a' and b' so that $f(c) = 0$. This new range is a quarter of the length of the original. If we keep doing this process the range will halve each time until we know that the zero is inside some tiny range of possible values. This process is called the bisection method.

Consider the following zero-finding example

Example 1.6.14

Show that the function $f(x) = x - 1 + \sin(\pi x/2)$ has a zero in $0 \leq x \leq 1$.

This question has been set up nicely to lead us towards using the IVT; we are already given a nice interval on which to look. In general we might have to test a few points and experiment a bit with a calculator before we can start narrowing down a range.

Let us start by testing the endpoints of the interval we are given

$$\begin{aligned} f(0) &= 0 - 1 + \sin(0) = -1 < 0 \\ f(1) &= 1 - 1 + \sin(\pi/2) = 1 > 0 \end{aligned}$$

So we know a point where f is positive and one where it is negative. So by the IVT there is a point in between where it is zero.

BUT in order to apply the IVT we have to show that the function is continuous, and we cannot simply write

it is continuous

We need to explain to the reader *why* it is continuous. That is — we have to prove it.

So to write up our answer we can put something like the following — keeping in mind we need to tell the reader what we are doing so they can follow along easily.

- We will use the IVT to prove that there is a zero in $[0, 1]$.
- First we must show that the function is continuous.
 - Since $x - 1$ is a polynomial it is continuous everywhere.
 - The function $\sin(\pi x/2)$ is a trigonometric function and is also continuous everywhere.
 - The sum of two continuous functions is also continuous, so $f(x)$ is continuous everywhere.
- Let $a = 0, b = 1$, then

$$\begin{aligned} f(0) &= 0 - 1 + \sin(0) = -1 < 0 \\ f(1) &= 1 - 1 + \sin(\pi/2) = 1 > 0 \end{aligned}$$

- The function is negative at $x = 0$ and positive at $x = 1$. Since the function is continuous we know there is a point $c \in [0, 1]$ so that $f(c) = 0$.

Notice that though we have not used full sentences in our explanation here, we are still using words. Your mathematics, unless it is very straight-forward computation, should contain words as well as symbols.

Example 1.6.14

The zero is actually located at about $x = 0.4053883559$.

The bisection method is really just the idea that we can keep repeating the above reasoning (with a calculator handy). Each iteration will tell us the location of the zero more precisely. The following example illustrates this.

Example 1.6.15

Use the bisection method to find a zero of

$$f(x) = x - 1 + \sin(\pi x/2)$$

that lies between 0 and 1.

So we start with the two points we worked out above:

- $a = 0, b = 1$ and

$$\begin{aligned} f(0) &= -1 \\ f(1) &= 1 \end{aligned}$$

- Test the point in the middle $x = \frac{0+1}{2} = 0.5$

$$f(0.5) = 0.2071067813 > 0$$

- So our new interval will be $[0, 0.5]$ since the function is negative at $x = 0$ and positive at $x = 0.5$

Repeat

- $a = 0, b = 0.5$ where $f(0) < 0$ and $f(0.5) > 0$.
- Test the point in the middle $x = \frac{0+0.5}{2} = 0.25$

$$f(0.25) = -0.3673165675 < 0$$

- So our new interval will be $[0.25, 0.5]$ since the function is negative at $x = 0.25$ and positive at $x = 0.5$

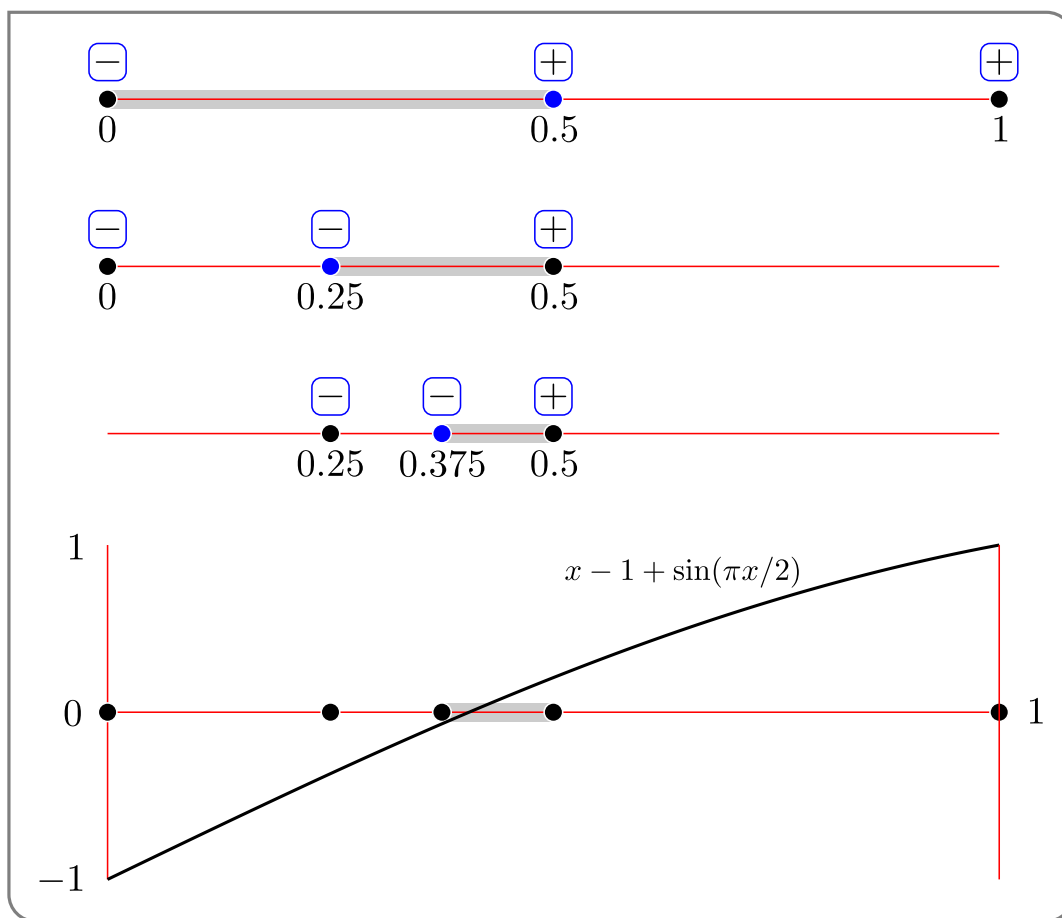
Repeat

- $a = 0.25, b = 0.5$ where $f(0.25) < 0$ and $f(0.5) > 0$.
- Test the point in the middle $x = \frac{0.25+0.5}{2} = 0.375$

$$f(0.375) = -0.0694297669 < 0$$

- So our new interval will be $[0.375, 0.5]$ since the function is negative at $x = 0.375$ and positive at $x = 0.5$

Below is an illustration of what we have observed so far together with a plot of the actual function.



And one final iteration:

- $a = 0.375, b = 0.5$ where $f(0.375) < 0$ and $f(0.5) > 0$.
- Test the point in the middle $x = \frac{0.375+0.5}{2} = 0.4375$

$$f(0.4375) = 0.0718932843 > 0$$

- So our new interval will be $[0.375, 0.4375]$ since the function is negative at $x = 0.375$ and positive at $x = 0.4375$

So without much work we know the location of a zero inside a range of length $0.0625 = 2^{-4}$. Each iteration will halve the length of the range and we keep going until we reach the precision we need, though it is much easier to program a computer to do it.

Example 1.6.15

1.7 ▲ (Optional) — Making the Informal a Little More Formal

As we noted above, the definition of limits that we have been working with was quite informal and not mathematically rigorous. In this (optional) section we will work to understand the rigorous definition of limits.

Here is the formal definition — we will work through it all very slowly and carefully afterwards, so do not panic.

Definition 1.7.1.

Let $a \in \mathbb{R}$ and let $f(x)$ be a function defined everywhere in a neighbourhood of a , except possibly at a . We say that

the limit as x approaches a of $f(x)$ is L

or equivalently

as x approaches a , $f(x)$ approaches L

and write

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if for every $\epsilon > 0$ there exists $\delta > 0$ so that

$$|f(x) - L| < \epsilon \text{ whenever } 0 < |x - a| < \delta$$

Note that an equivalent way of writing this very last statement is

$$\text{if } 0 < |x - a| < \delta \text{ then } |f(x) - L| < \epsilon.$$

1.5▲ Limits at Infinity

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Give a polynomial $f(x)$ with the property that both $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$ are (finite) real numbers.

Q[2]: Give a polynomial $f(x)$ that satisfies $\lim_{x \rightarrow \infty} f(x) \neq \lim_{x \rightarrow -\infty} f(x)$.

►► Stage 2

Q[3]: Evaluate $\lim_{x \rightarrow \infty} 2^{-x}$

Q[4]: Evaluate $\lim_{x \rightarrow \infty} 2^x$

Q[5]: Evaluate $\lim_{x \rightarrow -\infty} 2^x$

Q[6]: Evaluate $\lim_{x \rightarrow -\infty} \cos x$

Q[7]: Evaluate $\lim_{x \rightarrow \infty} x - 3x^5 + 100x^2$.

Q[8]: Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{3x^8 + 7x^4} + 10}{x^4 - 2x^2 + 1}$.

Q[9](*) : $\lim_{x \rightarrow \infty} \left[\sqrt{x^2 + 5x} - \sqrt{x^2 - x} \right]$

Q[10](*) : Evaluate $\lim_{x \rightarrow -\infty} \frac{3x}{\sqrt{4x^2 + x} - 2x}$.

Q[11](*) : Evaluate $\lim_{x \rightarrow -\infty} \frac{1 - x - x^2}{2x^2 - 7}$.

Q[12](*) : Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$

Q[13](*) : Evaluate $\lim_{x \rightarrow +\infty} \frac{5x^2 - 3x + 1}{3x^2 + x + 7}$.

Q[14](*) : Evaluate $\lim_{x \rightarrow +\infty} \frac{\sqrt{4x+2}}{3x+4}$.

Q[15](*) : Evaluate $\lim_{x \rightarrow +\infty} \frac{4x^3 + x}{7x^3 + x^2 - 2}$.

Q[16]: Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^2 + x} - \sqrt[4]{x^4 + 5}}{x + 1}$

Q[17](*) : Evaluate $\lim_{x \rightarrow +\infty} \frac{5x^2 + 10}{3x^3 + 2x^2 + x}$.

Q[18]: Evaluate $\lim_{x \rightarrow -\infty} \frac{x + 1}{\sqrt{x^2}}$.

Q[19]: Evaluate $\lim_{x \rightarrow \infty} \frac{x + 1}{\sqrt{x^2}}$

Q[20](*) : Find the limit $\lim_{x \rightarrow -\infty} \sin\left(\frac{\pi |x|}{2x}\right) + \frac{1}{x}$.

Q[21](*) : Evaluate $\lim_{x \rightarrow -\infty} \frac{3x + 5}{\sqrt{x^2 + 5} - x}$.

Q[22](*) : Evaluate $\lim_{x \rightarrow -\infty} \frac{5x + 7}{\sqrt{4x^2 + 15} - x}$

Q[23]: Evaluate $\lim_{x \rightarrow -\infty} \frac{3x^7 + x^5 - 15}{4x^2 + 32x}$.

Q[24](*) : Evaluate $\lim_{n \rightarrow \infty} \left(\sqrt{n^2 + 5n} - n \right)$.

Q[25]: Evaluate $\lim_{a \rightarrow 0^+} \frac{a^2 - \frac{1}{a}}{a - 1}$.

Q[26]: Evaluate $\lim_{x \rightarrow 3} \frac{2x + 8}{\frac{1}{x-3} + \frac{1}{x^2-9}}$.

► Stage 3

Q[27]: Give a rational function $f(x)$ with the properties that $\lim_{x \rightarrow \infty} f(x) \neq \lim_{x \rightarrow -\infty} f(x)$, and both limits are (finite) real numbers.

Q[28]: Suppose the concentration of a substance in your body t hours after injection is given by some formula $c(t)$, and $\lim_{t \rightarrow \infty} c(t) \neq 0$. What kind of substance might have been injected?

1.6▲ Continuity

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Give an example of a function (you can write a formula, or sketch a graph) that has infinitely many infinite discontinuities.

Q[2]: When I was born, I was less than one meter tall. Now, I am more than one meter tall. What is the conclusion of the Intermediate Value Theorem about my height?

Q[3]: Give an example (by sketch or formula) of a function $f(x)$, defined on the interval $[0, 2]$, with $f(0) = 0$, $f(2) = 2$, and $f(x)$ never equal to 1. Why does this not contradict the Intermediate Value Theorem?

Q[4]: Is the following a valid statement?

Suppose f is a continuous function over $[10, 20]$, $f(10) = 13$, and $f(20) = -13$. Then f has a zero between $x = 10$ and $x = 20$.

Q[5]: Is the following a valid statement?

Suppose f is a continuous function over $[10, 20]$, $f(10) = 13$, and $f(20) = -13$. Then $f(15) = 0$.

Q[6]: Is the following a valid statement?

Suppose f is a function over $[10, 20]$, $f(10) = 13$, and $f(20) = -13$, and f takes on every value between -13 and 13 . Then f is continuous.

Q[7]: Suppose $f(t)$ is continuous at $t = 5$. True or false: $t = 5$ is in the domain of $f(t)$.

Q[8]: Suppose $\lim_{t \rightarrow 5} f(t) = 17$, and suppose $f(t)$ is continuous at $t = 5$. True or false: $f(5) = 17$.

Q[9]: Suppose $\lim_{t \rightarrow 5} f(t) = 17$. True or false: $f(5) = 17$.

Q[10]: Suppose $f(x)$ and $g(x)$ are continuous at $x = 0$, and let $h(x) = \frac{xf(x)}{g^2(x) + 1}$. What is $\lim_{x \rightarrow 0^+} h(x)$?

► Stage 2

Q[11]: Find a constant k so that the function

$$a(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{when } x \neq 0 \\ k & \text{when } x = 0 \end{cases}$$

is continuous at $x = 0$.

Q[12]: Use the Intermediate Value Theorem to show that the function $f(x) = x^3 + x^2 + x + 1$ takes on the value 12345 at least once in its domain.

Q[13](*) : Describe all points for which the function is continuous: $f(x) = \frac{1}{x^2 - 1}$.

Q[14](*) : Describe all points for which this function is continuous: $f(x) = \frac{1}{\sqrt{x^2 - 1}}$.

Q[15](*) : Describe all points for which this function is continuous: $\frac{1}{\sqrt{1 + \cos(x)}}$.

Q[16](*) : Describe all points for which this function is continuous: $f(x) = \frac{1}{\sin x}$.

Q[17](*) : Find all values of c such that the following function is continuous at $x = c$:

$$f(x) = \begin{cases} 8 - cx & \text{if } x \leq c \\ x^2 & \text{if } x > c \end{cases}$$

Use the definition of continuity to justify your answer.

Q[18](*) : Find all values of c such that the following function is continuous everywhere:

$$f(x) = \begin{cases} x^2 + c & x \geq 0 \\ \cos cx & x < 0 \end{cases}$$

Use the definition of continuity to justify your answer.

Q[19](*) : Find all values of c such that the following function is continuous:

$$f(x) = \begin{cases} x^2 - 4 & \text{if } x < c \\ 3x & \text{if } x \geq c. \end{cases}$$

Use the definition of continuity to justify your answer.

Q[20](*) : Find all values of c such that the following function is continuous:

$$f(x) = \begin{cases} 6 - cx & \text{if } x \leq 2c \\ x^2 & \text{if } x > 2c \end{cases}$$

Use the definition of continuity to justify your answer.

► Stage 3

Q[21]: Show that there exists at least one real number x satisfying $\sin x = x - 1$

Q[22](*) : Show that there exists at least one real number c such that $3^c = c^2$.

Q[23](*) : Show that there exists at least one real number c such that $2 \tan(c) = c + 1$.

Q[24](*) : Show that there exists at least one real number c such that $\sqrt{\cos(\pi c)} = \sin(2\pi c) + \frac{1}{2}$.

Q[25](*) : Show that there exists at least one real number c such that $\frac{1}{(\cos \pi c)^2} = c + \frac{3}{2}$.

Q[26]: Use the intermediate value theorem to find an interval of length one containing a root of $f(x) = x^7 - 15x^6 + 9x^2 - 18x + 15$.

Q[27]: Use the intermediate value theorem to give a decimal approximation of $\sqrt[3]{7}$ that is correct to at least two decimal places. You may use a calculator, but only to add, subtract, multiply, and divide.

Q[28]: Suppose $f(x)$ and $g(x)$ are functions that are continuous over the interval $[a, b]$, with $f(a) \leq g(a)$ and $g(b) \leq f(b)$. Show that there exists some $c \in [a, b]$ with $f(c) = g(c)$.

−4. For these values, $f(x) = x^3 + 8x^2 + 16x$. So,

$$\lim_{x \rightarrow -4^-} f(x) = \lim_{x \rightarrow -4^-} (x^3 + 8x^2 + 16x) = (-4)^3 + 8(-4)^2 + 16(-4) = 0$$

Note that, because $x^3 + 8x^2 + 16x$ is a polynomial, we can evaluate the limit by directly substituting in $x = -4$.

(b) When we evaluate $\lim_{x \rightarrow -4^+} f(x)$, we only consider values of x that are greater than -4 .

For these values,

$$f(x) = \frac{x^2 + 8x + 16}{x^2 + 30x - 4}$$

So,

$$\lim_{x \rightarrow -4^+} f(x) = \lim_{x \rightarrow -4^+} \frac{x^2 + 8x + 16}{x^2 + 30x - 4}$$

This is a rational function, and $x = -4$ is in its domain (we aren't doing anything suspect, like dividing by 0), so again we can directly substitute $x = -4$ to evaluate the limit:

$$= \frac{(-4)^2 + 8(-4) + 16}{(-4)^2 + 30(-4) - 4} = \frac{0}{-108} = 0$$

(c) Since $\lim_{x \rightarrow -4^-} f(x) = \lim_{x \rightarrow -4^+} f(x) = 0$, we conclude $\lim_{x \rightarrow -4} f(x) = 0$.

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S-1: Any polynomial of degree one or higher will go to ∞ or $-\infty$ as x goes to ∞ . So, we need a polynomial of degree 0—that is, $f(x)$ is a constant. One possible answer is $f(x) = 1$.

S-2: This will be the case for any polynomial of *odd* degree. For instance, $f(x) = x$.

Many answers are possible: also $f(x) = x^{15} - 32x^2 + 9$ satisfies $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$.

S-3: $\lim_{x \rightarrow \infty} 2^{-x} = \lim_{x \rightarrow \infty} \frac{1}{2^x} = 0$

S-4: As x gets larger and larger, 2^x grows without bound. (For integer values of x , you can imagine multiplying 2 by itself more and more times.) So, $\lim_{x \rightarrow \infty} 2^x = \infty$.

S-5: Write $X = -x$. As x becomes more and more negative, X becomes more and more positive. From Question [4](#), we know that 2^X grows without bound as X gets larger and

larger. Since $2^x = 2^{-(-x)} = 2^{-X} = \frac{1}{2^X}$, as we let x become a huge negative number, we are in effect dividing by a huge positive number; hence $\lim_{x \rightarrow -\infty} 2^x = 0$.

A more formulaic way to describe the above is this: $\lim_{x \rightarrow -\infty} 2^x = \lim_{X \rightarrow \infty} 2^{-X} = \lim_{X \rightarrow \infty} \frac{1}{2^X} = 0$.

S-6: There is no single number that $\cos x$ approaches as x becomes more and more strongly negative: as x grows in the negative direction, the function oscillates between -1 and $+1$, never settling close to one particular number. So, this limit does not exist.

S-7: The highest-order term in this polynomial is $-3x^5$, so this dominates the function's behaviour as x goes to infinity. More formally:

$$\begin{aligned} \lim_{x \rightarrow \infty} (x - 3x^5 + 100x^2) &= \lim_{x \rightarrow \infty} -3x^5 \left(1 - \frac{1}{3x^4} - \frac{100}{3x^3} \right) \\ &= \lim_{x \rightarrow \infty} -3x^5 = -\infty \end{aligned}$$

because

$$\lim_{x \rightarrow \infty} \left(1 - \frac{1}{3x^4} - \frac{100}{3x^3} \right) = 1 - 0 - 0 = 1.$$

S-8: Our standard trick is to factor out the highest power of x in the denominator: x^4 . We just have to be a little careful with the square root. Since we are taking the limit as x goes to positive infinity, we have positive x -values, so $\sqrt{x^2} = x$ and $\sqrt{x^8} = x^4$.

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{3x^8 + 7x^4} + 10}{x^4 - 2x^2 + 1} &= \lim_{x \rightarrow \infty} \frac{\sqrt{x^8(3 + \frac{7}{x^4})} + 10}{x^4(1 - \frac{2}{x^2} + \frac{1}{x^4})} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{x^8} \sqrt{3 + \frac{7}{x^4}} + 10}{x^4(1 - \frac{2}{x^2} + \frac{1}{x^4})} \\ &= \lim_{x \rightarrow \infty} \frac{x^4 \sqrt{3 + \frac{7}{x^4}} + 10}{x^4(1 - \frac{2}{x^2} + \frac{1}{x^4})} \\ &= \lim_{x \rightarrow \infty} \frac{x^4 \left(\sqrt{3 + \frac{7}{x^4}} + \frac{10}{x^4} \right)}{x^4(1 - \frac{2}{x^2} + \frac{1}{x^4})} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{3 + \frac{7}{x^4}} + \frac{10}{x^4}}{1 - \frac{2}{x^2} + \frac{1}{x^4}} \\ &= \frac{\sqrt{3+0}+0}{1-0+0} = \sqrt{3} \end{aligned}$$

S-9: We have two terms, each getting extremely large. It's unclear at first what happens when we subtract them. To get this equation into another form, we multiply and divide by the conjugate, $\sqrt{x^2 + 5x} + \sqrt{x^2 - x}$.

$$\begin{aligned}\lim_{x \rightarrow \infty} [\sqrt{x^2 + 5x} - \sqrt{x^2 - x}] &= \lim_{x \rightarrow \infty} \left[\frac{(\sqrt{x^2 + 5x} - \sqrt{x^2 - x})(\sqrt{x^2 + 5x} + \sqrt{x^2 - x})}{\sqrt{x^2 + 5x} + \sqrt{x^2 - x}} \right] \\ &= \lim_{x \rightarrow \infty} \frac{(x^2 + 5x) - (x^2 - x)}{\sqrt{x^2 + 5x} + \sqrt{x^2 - x}} \\ &= \lim_{x \rightarrow \infty} \frac{6x}{\sqrt{x^2 + 5x} + \sqrt{x^2 - x}}\end{aligned}$$

Now we divide the numerator and denominator by x . In the case of the denominator, since $x > 0$, $x = \sqrt{x^2}$.

$$\begin{aligned}&= \lim_{x \rightarrow \infty} \frac{6(x)}{\sqrt{x^2}\sqrt{1 + \frac{5}{x}} + \sqrt{x^2}\sqrt{1 - \frac{1}{x}}} \\ &= \lim_{x \rightarrow \infty} \frac{6(x)}{(x)\sqrt{1 + \frac{5}{x}} + (x)\sqrt{1 - \frac{1}{x}}} \\ &= \lim_{x \rightarrow \infty} \frac{6}{\sqrt{1 + \frac{5}{x}} + \sqrt{1 - \frac{1}{x}}} \\ &= \frac{6}{\sqrt{1 + 0} + \sqrt{1 - 0}} = 3\end{aligned}$$

S-10: Note that for large negative x , the first term in the denominator $\sqrt{4x^2 + x} \approx \sqrt{4x^2} = |2x| = -2x$ not $+2x$. A good way to avoid incorrectly computing $\sqrt{x^2}$ when x is negative is to define $y = -x$ and express everything in terms of y . That's what we'll do.

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{3x}{\sqrt{4x^2 + x} - 2x} &= \lim_{y \rightarrow +\infty} \frac{-3y}{\sqrt{4y^2 - y} + 2y} \\ &= \lim_{y \rightarrow +\infty} \frac{-3y}{y\sqrt{4 - \frac{1}{y}} + 2y} \\ &= \lim_{y \rightarrow +\infty} \frac{-3}{\sqrt{4 - \frac{1}{y}} + 2} \\ &= \frac{-3}{\sqrt{4 - 0} + 2} \quad \text{since } 1/y \rightarrow 0 \text{ as } y \rightarrow +\infty \\ &= -\frac{3}{4}\end{aligned}$$

S-11: The highest power of x in the denominator is x^2 , so we divide the numerator and denominator by x^2 :

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{1 - x - x^2}{2x^2 - 7} &= \lim_{x \rightarrow -\infty} \frac{1/x^2 - 1/x - 1}{2 - 7/x^2} \\ &= \frac{0 - 0 - 1}{2 - 0} = -\frac{1}{2}\end{aligned}$$

S-12:

$$\begin{aligned}\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x} - x)(\sqrt{x^2 + x} + x)}{\sqrt{x^2 + x} + x} = \lim_{x \rightarrow \infty} \frac{(x^2 + x) - x^2}{\sqrt{x^2 + x} + x} \\ &= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + x} + x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} = \frac{1}{2}\end{aligned}$$

S-13: We have, after dividing both numerator and denominator by x^2 (which is the highest power of the denominator) that

$$\frac{5x^2 - 3x + 1}{3x^2 + x + 7} = \frac{5 - \frac{3}{x} + \frac{1}{x^2}}{3 + \frac{1}{x} + \frac{7}{x^2}}.$$

Since $1/x \rightarrow 0$ and also $1/x^2 \rightarrow 0$ as $x \rightarrow +\infty$, we conclude that

$$\lim_{x \rightarrow +\infty} \frac{5x^2 - 3x + 1}{3x^2 + x + 7} = \frac{5}{3}.$$

S-14: We have, after dividing both numerator and denominator by x (which is the highest power of the denominator) that

$$\frac{\sqrt{4x+2}}{3x+4} = \frac{\sqrt{\frac{4}{x} + \frac{2}{x^2}}}{3 + \frac{4}{x}}.$$

Since $1/x \rightarrow 0$ and also $1/x^2 \rightarrow 0$ as $x \rightarrow +\infty$, we conclude that

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x+2}}{3x+4} = \frac{0}{3} = 0.$$

S-15: The dominant terms in the numerator and denominator have order x^3 . Taking out that common factor we get

$$\frac{4x^3 + x}{7x^3 + x^2 - 2} = \frac{4 + \frac{1}{x^2}}{7 + \frac{1}{x} - \frac{2}{x^3}}.$$

Since $1/x^a \rightarrow 0$ as $x \rightarrow +\infty$ (for $a > 0$), we conclude that

$$\lim_{x \rightarrow +\infty} \frac{4x^3 + x}{7x^3 + x^2 - 2} = \frac{4}{7}.$$

S-16:

• Solution 1

We want to factor out x , the highest power in the denominator. Since our limit only sees negative values of x , we must remember that $\sqrt[4]{x^4} = |x| = -x$, although $\sqrt[3]{x^3} = x$.

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^2 + x} - \sqrt[4]{x^4 + 5}}{x + 1} &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^3(\frac{1}{x} + \frac{1}{x^2})} - \sqrt[4]{x^4(1 + \frac{5}{x^4})}}{x(1 + \frac{1}{x})} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^3} \sqrt[3]{\frac{1}{x} + \frac{1}{x^2}} - \sqrt[4]{x^4} \sqrt[4]{1 + \frac{5}{x^4}}}{x(1 + \frac{1}{x})} \\ &= \lim_{x \rightarrow -\infty} \frac{x \sqrt[3]{\frac{1}{x} + \frac{1}{x^2}} - (-x) \sqrt[4]{1 + \frac{5}{x^4}}}{x(1 + \frac{1}{x})} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{\frac{1}{x} + \frac{1}{x^2}} + \sqrt[4]{1 + \frac{5}{x^4}}}{1 + \frac{1}{x}} \\ &= \frac{\sqrt[3]{0 + 0} + \sqrt[4]{1 + 0}}{1 + 0} = 1 \end{aligned}$$

• Solution 2

Alternately, we can use the transformation $\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow \infty} f(-x)$. Then we only look at positive values of x , so roots behave nicely: $\sqrt[4]{x^4} = |x| = x$.

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^2 + x} - \sqrt[4]{x^4 + 5}}{x + 1} &= \lim_{x \rightarrow \infty} \frac{\sqrt[3]{(-x)^2 - x} - \sqrt[4]{(-x)^4 + 5}}{-x + 1} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^2 - x} - \sqrt[4]{x^4 + 5}}{-x + 1} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^3} \sqrt[3]{\frac{1}{x} - \frac{1}{x^2}} - \sqrt[4]{x^4} \sqrt[4]{1 + \frac{5}{x^4}}}{x(-1 + \frac{1}{x})} \\ &= \lim_{x \rightarrow \infty} \frac{x \sqrt[3]{\frac{1}{x} - \frac{1}{x^2}} - x \sqrt[4]{1 + \frac{5}{x^4}}}{x(-1 + \frac{1}{x})} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt[3]{\frac{1}{x} - \frac{1}{x^2}} - \sqrt[4]{1 + \frac{5}{x^4}}}{-1 + \frac{1}{x}} \\ &= \frac{\sqrt[3]{0 - 0} - \sqrt[4]{1 + 0}}{-1 + 0} = \frac{-1}{-1} = 1 \end{aligned}$$

S-17: We have, after dividing both numerator and denominator by x^3 (which is the highest power of the denominator) that:

$$\lim_{x \rightarrow \infty} \frac{5x^2 + 10}{3x^3 + 2x^2 + x} = \lim_{x \rightarrow \infty} \frac{\frac{5}{x} + \frac{10}{x^3}}{3 + \frac{2}{x} + \frac{1}{x^2}} = \frac{0}{3} = 0.$$

S-18: Since we only consider negative values of x , $\sqrt{x^2} = |x| = -x$.

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{x+1}{\sqrt{x^2}} &= \lim_{x \rightarrow -\infty} \frac{x+1}{-x} \\ &= \lim_{x \rightarrow -\infty} \frac{x}{-x} + \frac{1}{-x} \\ &= \lim_{x \rightarrow -\infty} -1 - \frac{1}{x} \\ &= -1 \end{aligned}$$

S-19: Since we only consider positive values of x , $\sqrt{x^2} = |x| = x$.

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x+1}{\sqrt{x^2}} &= \lim_{x \rightarrow \infty} \frac{x+1}{x} \\ &= \lim_{x \rightarrow \infty} 1 + \frac{1}{x} = 1 + 0 = 1 \end{aligned}$$

S-20: When $x < 0$, $|x| = -x$ and so $\lim_{x \rightarrow \infty} \sin\left(\frac{\pi}{2} \cdot \frac{|x|}{x}\right) + \frac{1}{x} = \sin(-\pi/2) = -1$.

S-21: We divide both the numerator and the denominator by the highest power of x in the denominator, which is x . Since $x < 0$, we have $\sqrt{x^2} = |x| = -x$, so that

$$\frac{\sqrt{x^2+5}}{x} = -\sqrt{\frac{x^2+5}{x^2}} = -\sqrt{1+\frac{5}{x^2}}.$$

Since $1/x \rightarrow 0$ and also $1/x^2 \rightarrow 0$ as $x \rightarrow -\infty$, we conclude that

$$\lim_{x \rightarrow -\infty} \frac{3x+5}{\sqrt{x^2+5}-x} = \lim_{x \rightarrow -\infty} \frac{3+\frac{5}{x}}{-\sqrt{1+\frac{5}{x^2}}-1} = \frac{3}{-1-1} = -\frac{3}{2}.$$

S-22: We divide both the numerator and the denominator by the highest power of x in the denominator, which is x . Since $x < 0$, we have $\sqrt{x^2} = |x| = -x$, so that

$$\frac{\sqrt{4x^2 + 15}}{x} = \frac{\sqrt{4x^2 + 15}}{-\sqrt{x^2}} = -\sqrt{\frac{4x^2 + 15}{x^2}} = -\sqrt{4 + \frac{15}{x^2}}.$$

Since $1/x \rightarrow 0$ and also $1/x^2 \rightarrow 0$ as $x \rightarrow -\infty$, we conclude that

$$\lim_{x \rightarrow -\infty} \frac{5x + 7}{\sqrt{4x^2 + 15} - x} = \lim_{x \rightarrow -\infty} \frac{5 + \frac{7}{x}}{-\sqrt{4 + \frac{15}{x^2}} - 1} = \frac{5}{-2 - 1} = -\frac{5}{3}.$$

S-23:

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{3x^7 + x^5 - 15}{4x^2 + 32x} &= \lim_{x \rightarrow -\infty} \frac{x^2(3x^5 + x^3 - \frac{15}{x^2})}{x^2(4 + \frac{32}{x})} \\ &= \lim_{x \rightarrow -\infty} \frac{3x^5 + x^3 - \frac{15}{x^2}}{4 + \frac{32}{x}} \\ &= \lim_{x \rightarrow +\infty} \frac{3(-x)^5 + (-x)^3 - \frac{15}{(-x)^2}}{4 + \frac{32}{-x}} \\ &= \lim_{x \rightarrow +\infty} \frac{-3x^5 - x^3 - \frac{15}{x^2}}{4 - \frac{32}{x}} \\ &= -\infty \end{aligned}$$

S-24: We multiply and divide the expression by its conjugate, $(\sqrt{n^2 + 5n} + n)$.

$$\begin{aligned} \lim_{n \rightarrow \infty} (\sqrt{n^2 + 5n} - n) &= \lim_{n \rightarrow \infty} (\sqrt{n^2 + 5n} - n) \left(\frac{\sqrt{n^2 + 5n} + n}{\sqrt{n^2 + 5n} + n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{(n^2 + 5n) - n^2}{\sqrt{n^2 + 5n} + n} \\ &= \lim_{n \rightarrow \infty} \frac{5n}{\sqrt{n^2 + 5n} + n} \\ &= \lim_{n \rightarrow \infty} \frac{5 \cdot n}{\sqrt{n^2} \sqrt{1 + \frac{5}{n}} + n} \end{aligned}$$

Since $n > 0$, we can simplify $\sqrt{n^2} = n$.

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} \frac{5 \cdot n}{n\sqrt{1 + \frac{5}{n}} + n} \\
 &= \lim_{n \rightarrow \infty} \frac{5}{\sqrt{1 + \frac{5}{n}} + 1} \\
 &= \frac{5}{\sqrt{1 + 0} + 1} = \frac{5}{2}
 \end{aligned}$$

S-25:

• **Solution 1:**

When a approaches 0 from the right, the numerator approaches negative infinity, and the denominator approaches -1 . So, $\lim_{a \rightarrow 0^+} \frac{a^2 - \frac{1}{a}}{a - 1} = \infty$.

More precisely, using Theorem 1.5.9:

$$\lim_{a \rightarrow 0^+} \frac{1}{a} = +\infty$$

$$\text{Also, } \lim_{a \rightarrow 0^+} a^2 = 0$$

$$\text{So, using Theorem 1.5.9, } \lim_{a \rightarrow 0^+} a^2 - \frac{1}{a} = -\infty$$

$$\text{Furthermore, } \lim_{a \rightarrow 0^+} a - 1 = -1$$

$$\text{So, using our theorem, } \lim_{a \rightarrow 0^+} \frac{a^2 - \frac{1}{a}}{a - 1} = \infty$$

• **Solution 2:**

Since $a = 0$ is not in the domain of our function, a reasonable impulse is to simplify.

$$\frac{a^2 - \frac{1}{a}}{a - 1} \left(\frac{a}{a} \right) = \frac{a^3 - 1}{a(a - 1)} = \frac{(a - 1)(a^2 + a + 1)}{a(a - 1)}$$

So,

$$\begin{aligned}
 \lim_{a \rightarrow 0^+} \frac{a^2 - \frac{1}{a}}{a - 1} &= \lim_{a \rightarrow 0^+} \frac{(a - 1)(a^2 + a + 1)}{a(a - 1)} \\
 &= \lim_{a \rightarrow 0^+} \frac{a^2 + a + 1}{a} \\
 &= \lim_{a \rightarrow 0^+} a + 1 + \frac{1}{a} = \infty
 \end{aligned}$$

S-26:

Since $x = 3$ is not in the domain of the function, we simplify, hoping we can cancel a problematic term.

$$\begin{aligned}\lim_{x \rightarrow 3} \frac{2x + 8}{\frac{1}{x-3} + \frac{1}{x^2-9}} &= \lim_{x \rightarrow 3} \frac{2x + 8}{\frac{x+3}{x^2-9} + \frac{1}{x^2-9}} \\ &= \lim_{x \rightarrow 3} \frac{2x + 8}{\frac{x+4}{x^2-9}} \\ &= \lim_{x \rightarrow 3} \frac{(2x + 8)(x^2 - 9)}{x + 4} = 0\end{aligned}$$

S-27: First, we need a rational function whose limit at infinity is a real number. This means that the degree of the bottom is greater than or equal to the degree of the top. There are two cases: the denominator has higher degree than the numerator, or the denominator has the same degree as the numerator.

If the denominator has higher degree than the numerator, then

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = 0, \text{ so the limits are equal—not what we're looking for.}$$

If the denominator has the same degree as the numerator, then the limit as x goes to $\pm\infty$ is the ratio of the leading terms: again, the limits are equal. So no rational function exists as described.

S-28: The amount of the substance that will linger long-term is some positive number—the substance will stick around. One example of a substance that does this is the ink in a tattoo. (If the injection was of medicine, probably it will be metabolized, and $\lim_{t \rightarrow \infty} c(t) = 0$.)

Remark: it actually doesn't make much sense to let t go to infinity: after a few million hours, you won't even have a body, so what is $c(t)$ measuring? Often when we use formulas in the real world, there is an understanding that they are only good for some fixed range. We often use the limit as t goes to infinity as a stand-in for the function's long-term behaviour.

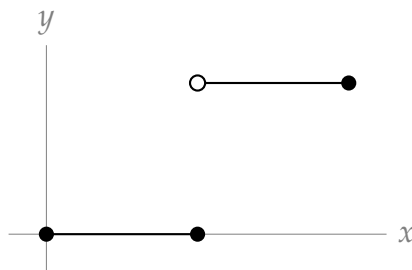
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S-1: Many answers are possible; the tangent function behaves like this.

S-2: If we let f be my height, the time of my birth is a , and now is b , then we know that $f(a) \leq 1 \leq f(b)$. It is reasonable to assume that my height is a continuous function. So by the IVT, there is some value c between a and b where $f(c) = 1$. That is, there is some time (we called it c) between my birth and today when I was exactly one meter tall.

Notice the IVT does not say precisely what day I was one meter tall; it only guarantees that such a day occurred between my birth and today.

S-3: One example is $f(x) = \begin{cases} 0 & \text{when } 0 \leq x \leq 1 \\ 2 & \text{when } 1 < x \leq 2 \end{cases}$. The IVT only guarantees $f(c) = 1$ for some c in $[0, 2]$ when f is *continuous* over $[0, 2]$. If f is not continuous, the IVT says nothing.

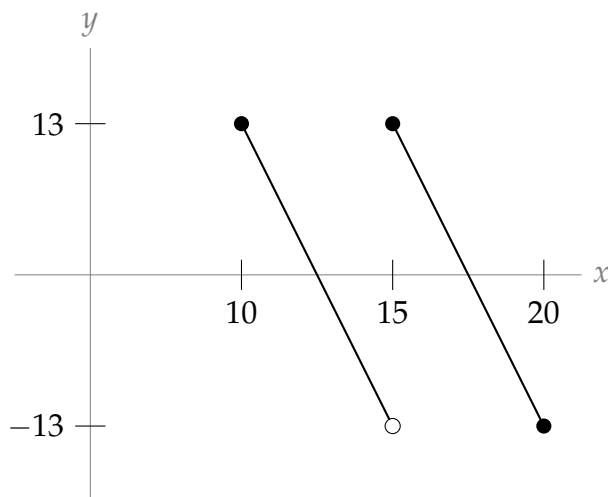


S-4: Yes. This is a straightforward application of IVT.

S-5: No. IVT says that $f(x) = 0$ for some x between 10 and 20, but it doesn't have to be *exactly half way* between.

S-6: No. IVT says nothing about functions that are not guaranteed to be continuous at the outset. It's quite easy to construct a function that is as described, but not continuous. For example, the function pictured below, whose equation format is somewhat less

enlightening than its graph: $f(x) = \begin{cases} -\frac{26}{5}x + 65, & 10 \leq x < 15 \\ -\frac{26}{5}x + 91, & 15 \leq x \leq 20 \end{cases}$.

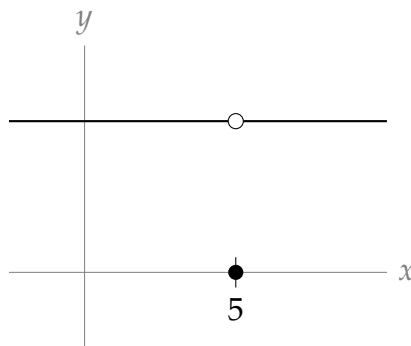


S-7: True. Since $f(t)$ is continuous at $t = 5$, that means $\lim_{t \rightarrow 5} f(t) = f(5)$. For that to be true, $f(5)$ must exist — that is, 5 is in the domain of $f(x)$.

S-8: True. Using the definition of continuity, $\lim_{t \rightarrow 5} f(t) = f(5) = 17$.

S-9: In general, false. If $f(t)$ is continuous at $t = 5$, then $f(5) = 17$; if $f(t)$ is discontinuous at $t = 5$, then $f(5)$ either does not exist, or is a number other than 17.

An example of a function with $\lim_{t \rightarrow 5} f(t) = 17 \neq f(5)$ is $f(t) = \begin{cases} 17 & , t \neq 5 \\ 0 & , t = 5 \end{cases}$, shown below.



S-10: Since $f(x)$ and $g(x)$ are continuous at zero, and since $g^2(x) + 1$ must be nonzero, then $h(x)$ is continuous at 0 as well. According to the definition of continuity, then $\lim_{x \rightarrow 0} h(x)$ exists and is equal to $h(0) = \frac{0f(0)}{g^2(0)+1} = 0$.

Since the limit $\lim_{x \rightarrow 0} h(x)$ exists and is equal to zero, also the one-sided limit $\lim_{x \rightarrow 0^+} h(x)$ exists and is equal to zero.

S-11: Using the definition of continuity, we need $k = \lim_{x \rightarrow 0} f(x)$. Since the limit is blind to what actually happens to $f(x)$ at $x = 0$, this is equivalent to $k = \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$. So if we find the limit, we solve the problem.

As x gets small, $\sin\left(\frac{1}{x}\right)$ goes a little crazy (see example 1.3.5), so let's get rid of it by using the Squeeze Theorem. We can bound the function above and below, but we should be a little careful about whether we're going from the left or the right. The reason we need to worry about direction is illustrated with the following observation: If $a \leq b$ and $x > 0$, then $xa \leq xb$. (For example, plug in $x = 1, a = 2, b = 3$.) But if $a \leq b$ and $x < 0$, then $xa \geq xb$. (For example, plug in $x = -1, a = 2, b = 3$.) So first, let's find $\lim_{x \rightarrow 0^-} \sin\left(\frac{1}{x}\right)$. When $x < 0$,

$$1 \geq \sin\left(\frac{1}{x}\right) \geq -1$$

$$\text{so: } x(1) \leq x \sin\left(\frac{1}{x}\right) \leq x(-1)$$

and $\lim_{x \rightarrow 0^-} x = \lim_{x \rightarrow 0^-} -x = 0$, so by the Squeeze Theorem, also $\lim_{x \rightarrow 0^-} x \sin\left(\frac{1}{x}\right) = 0$.

Now, let's find $\lim_{x \rightarrow 0^+} x \sin\left(\frac{1}{x}\right)$. When $x > 0$,

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

$$\text{so: } x(-1) \leq x \sin\left(\frac{1}{x}\right) \leq x(1)$$

and $\lim_{x \rightarrow 0^+} x = \lim_{x \rightarrow 0^+} -x = 0$, so by the Squeeze Theorem, also $\lim_{x \rightarrow 0^+} x \sin\left(\frac{1}{x}\right) = 0$.

Since the limits from the left and right agree, we conclude $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$, so when $k = 0$, the function is continuous at $x = 0$.

S-12: Since f is a polynomial, it is continuous over all real numbers. $f(0) = 1 < 12345$ and $f(12345) = 12345^3 + 12345^2 + 12345 + 1 > 12345$ (since all terms are positive). So by the IVT, $f(c) = 12345$ for some c between 0 and 12345.

S-13: $f(x)$ is a rational function and so is continuous except when its denominator is zero. That is, except when $x = 1$ and $x = -1$.

S-14: The function is continuous when $x^2 - 1 > 0$, i.e. $(x - 1)(x + 1) > 0$, which yields the intervals $(-\infty, -1) \cup (1, +\infty)$.

S-15: The function $1/\sqrt{x}$ is continuous on $(0, +\infty)$ and the function $\cos(x) + 1$ is continuous everywhere.

So $1/\sqrt{\cos(x) + 1}$ is continuous except when $\cos x = -1$. This happens when x is an odd multiple of π . Hence the function is continuous except at $x = \pm\pi, \pm3\pi, \pm5\pi, \dots$

S-16: The function is continuous when $\sin(x) \neq 0$. That is, when x is not an integer multiple of π .

S-17: The function is continuous for $x \neq c$ since each of those two branches are polynomials. So, the only question is whether the function is continuous at $x = c$; for this we need

$$\lim_{x \rightarrow c^-} f(x) = f(c) = \lim_{x \rightarrow c^+} f(x).$$

We compute

$$\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^-} 8 - cx = 8 - c^2;$$

$$f(c) = 8 - c \cdot c = 8 - c^2 \text{ and}$$

$$\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^+} x^2 = c^2.$$

So, we need $8 - c^2 = c^2$, which yields $c^2 = 4$, i.e. $c = -2$ or $c = 2$.

S-18: The function is continuous for $x \neq 0$ since $x^2 + c$ and $\cos cx$ are continuous everywhere. It remains to check continuity at $x = 0$. To do this we must check that the following three are equal.

$$\begin{aligned}\lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} x^2 + c = c \\ f(0) &= c \\ \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} \cos cx = \cos 0 = 1\end{aligned}$$

Hence when $c = 1$ we have the limits agree.

S-19: The function is continuous for $x \neq c$ since each of those two branches are defined by polynomials. Thus, the only question is whether the function is continuous at $x = c$. Furthermore,

$$\lim_{x \rightarrow c^-} f(x) = c^2 - 4$$

and

$$\lim_{x \rightarrow c^+} f(x) = f(c) = 3c.$$

For continuity we need both limits and the value to agree, so f is continuous if and only if $c^2 - 4 = 3c$, that is if and only if

$$c^2 - 3c - 4 = 0.$$

Factoring this as $(c - 4)(c + 1) = 0$ yields $c = -1$ or $c = +4$.

S-20: The function is continuous for $x \neq 2c$ since each of those two branches are polynomials. So, the only question is whether the function is continuous at $x = 2c$; for this we need

$$\lim_{x \rightarrow 2c^-} f(x) = f(2c) = \lim_{x \rightarrow 2c^+} f(x).$$

We compute

$$\lim_{x \rightarrow 2c^-} f(x) = \lim_{x \rightarrow 2c^-} 6 - cx = 6 - 2c^2;$$

$$f(2c) = 6 - c \cdot 2c = 6 - 2c^2 \text{ and}$$

$$\lim_{x \rightarrow 2c^+} f(x) = \lim_{x \rightarrow 2c^+} x^2 = 4c^2.$$

So, we need $6 - 2c^2 = 4c^2$, which yields $c^2 = 1$, i.e. $c = -1$ or $c = 1$.

S-21: This isn't the kind of equality that we can just solve; we'll need a trick, and that trick is the IVT. The general idea is to show that $\sin x$ is somewhere bigger, and somewhere smaller, than $x - 1$. However, since the IVT can only show us that a function is equal to a constant, we need to slightly adjust our language. Showing $\sin x = x - 1$ is equivalent to showing $\sin x - x + 1 = 0$, so let $f(x) = \sin x - x + 1$, and let's show that it has a real root.

First, we need to note that $f(x)$ is continuous (otherwise we can't use the IVT). Now, we need to find a value of x for which it is positive, and for which it's negative. By checking

a few values, we find $f(0)$ is positive, and $f(100)$ is negative. So, by the IVT, there exists a value of x (between 0 and 100) for which $f(x) = 0$. Therefore, there exists a value of x for which $\sin x = x - 1$.

S-22: We let $f(x) = 3^x - x^2$. Then $f(x)$ is a continuous function, since both 3^x and x^2 are continuous for all real numbers.

We find a value a such that $f(a) > 0$. We observe immediately that $a = 0$ works since

$$f(0) = 3^0 - 0 = 1 > 0.$$

We find a value b such that $f(b) < 0$. We see that $b = -1$ works since

$$f(-1) = \frac{1}{3} - 1 < 0.$$

So, because $f(x)$ is continuous on $(-\infty, \infty)$ and $f(0) > 0$ while $f(-1) < 0$, then the Intermediate Value Theorem guarantees the existence of a real number c in the interval $(-1, 0)$ such that $f(c) = 0$.

S-23: We let $f(x) = 2 \tan(x) - x - 1$. Then $f(x)$ is a continuous function on the interval $(-\pi/2, \pi/2)$ since $\tan(x) = \sin(x)/\cos(x)$ is continuous on this interval, while $x + 1$ is a polynomial and therefore continuous for all real numbers.

We find a value $a \in (-\pi/2, \pi/2)$ such that $f(a) < 0$. We observe immediately that $a = 0$ works since

$$f(0) = 2 \tan(0) - 0 - 1 = 0 - 1 = -1 < 0.$$

We find a value $b \in (-\pi/2, \pi/2)$ such that $f(b) > 0$. We see that $b = \pi/4$ works since

$$f(\pi/4) = 2 \tan(\pi/4) - \pi/4 - 1 = 2 - \pi/4 - 1 = 1 - \pi/4 = (4 - \pi)/4 > 0,$$

because $3 < \pi < 4$.

So, because $f(x)$ is continuous on $[0, \pi/4]$ and $f(0) < 0$ while $f(\pi/4) > 0$, then the Intermediate Value Theorem guarantees the existence of a real number $c \in (0, \pi/4)$ such that $f(c) = 0$.

S-24: Let $f(x) = \sqrt{\cos(\pi x)} - \sin(2\pi x) - 1/2$. This function is continuous provided $\cos(\pi x) \geq 0$. This is true for $0 \leq x \leq \frac{1}{2}$.

Now f takes positive values on $[0, 1/2]$:

$$f(0) = \sqrt{\cos(0)} - \sin(0) - 1/2 = \sqrt{1} - 1/2 = 1/2.$$

And f takes negative values on $[0, 1/2]$:

$$f(1/2) = \sqrt{\cos(\pi/2)} - \sin(\pi) - 1/2 = 0 - 0 - 1/2 = -1/2$$

(Notice that $f(1/3) = (\sqrt{2} - \sqrt{3})/2 - 1/2$ also works)

So, because $f(x)$ is continuous on $[0, 1/2)$ and $f(0) > 0$ while $f(1/2) < 0$, then the Intermediate Value Theorem guarantees the existence of a real number $c \in (0, 1/2)$ such that $f(c) = 0$.

S-25: We let $f(x) = \frac{1}{\cos^2(\pi x)} - x - \frac{3}{2}$. Then $f(x)$ is a continuous function on the interval $(-1/2, 1/2)$ since $\cos x$ is continuous everywhere and non-zero on that interval.

The function f takes negative values. For example, when $x = 0$:

$$f(0) = \frac{1}{\cos^2(0)} - 0 - \frac{3}{2} = 1 - \frac{3}{2} = -\frac{1}{2} < 0.$$

It also takes positive values, for instance when $x = 1/4$:

$$\begin{aligned} f(1/4) &= \frac{1}{(\cos \pi/4)^2} - \frac{1}{4} - \frac{3}{2} \\ &= \frac{1}{1/2} - \frac{1+6}{4} \\ &= 2 - 7/4 = 1/4 > 0. \end{aligned}$$

By the IVT there is c , $0 < c < 1/4$ such that $f(c) = 0$, in which case

$$\frac{1}{(\cos \pi c)^2} = c + \frac{3}{2}.$$

S-26: $f(x)$ is a polynomial, so it's continuous everywhere. If we can find values a and b so that $f(a) > 0$ and $f(b) < 0$, then by the IVT, there will exist some c in (a, b) where $f(c) = 0$; that is, there is a root in the interval $[a, b]$. Let's start plugging in easy values of x .

$f(0) = 15$, and $f(1) = 1 - 15 + 9 - 18 + 15 = -8$. Since 0 is between $f(0)$ and $f(1)$, and since f is continuous, by IVT there must be some x in $[0, 1]$ for which $f(x) = 0$: that is, there is some root in $[0, 1]$.

That was the easiest interval to find. If you keep playing around, you find two more. $f(-1) = 26$ (positive) and $f(-2) = -1001$ (negative), so there is a root in $[-2, -1]$.

The arithmetic is nasty, but there is also a root in $[14, 15]$.

This is an important trick. For high-degree polynomials, it is often impossible to get the exact values of the roots. Using the IVT, we can at least give a range where a root must be.

S-27: Let $f(x) = x^3$. Since f is a polynomial, it is continuous everywhere. If $f(a) < 7 < f(b)$, then $\sqrt[3]{7}$ is somewhere between a and b . If we can find a and b that satisfy these inequalities, and are very close together, that will give us a good approximation for $\sqrt[3]{7}$.